

Hellenic Complex Systems Laboratory

Quality: A Software Tool for Statistical Quality Control Design

Technical Report XX

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2022

Quality: A Software Tool for Statistical Quality Control Design

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DOI: [10.5281/zenodo.20278487](https://doi.org/10.5281/zenodo.20278487)

Abstract

This software tool can design and compare statistical single-value quality control rules to be applied to a measurement process to meet the user defined quality specifications.

Search Terms

quality control, quality control design, measurement, total allowable analytical error, fraction nonconforming, critical systematic error, critical random error, decision limits, probability for rejection, probability for false rejection

Introduction

Alternative quality control (QC) rules and procedures can be applied to a process to test statistically the null hypothesis that the process conforms to the quality specifications and consequently is in control, against the alternative hypothesis, that the process is out of control. A statistical type I error is committed when a true null hypothesis is rejected. We then have a false rejection of the run of the process. The probability of a type I error is called the probability for false rejection. A statistical type II error is committed when a false null hypothesis is accepted. We fail then to detect a significant change in the probability density function of a quality characteristic of the process. The probability of rejection of a false null hypothesis equals the probability of detection of the nonconformity of the process to the quality specifications.

The program *Quality* was developed to design and explore single-value QC rules to meet defined quality specifications.

The program

The program provides two modules:

Design

This module estimates various parameters of a measurement process, designs the single-value quality $S(\mathbf{x}; m, s, l_i)$ (see Notation) control rule to be applied. Then, it plots the probability density function (pdf) and the cumulative distribution function (cdf) of the control measurements, as well as the probability for rejection for the random error and systematic error of the QC rule (see Figures 1 and 2).

Compare

This module explores and compares pairs of alternative single-value statistical QC rules $S(\mathbf{x}; m, s, l_i)$ (see Notation) applied to tuples $\mathbf{x}$ of n_i control measurements, k_i of them distributed as $\mathcal{N}(m, s^2)$, and $n_i - k_i$

as $\mathcal{N}(0,1)$, plotting the probabilities for rejection $P(n_i, k_i, m, s; 0,1, l_i)$ of the two statistical QC rules, their difference, relative difference, and ratio (see Figures 3 and 4).

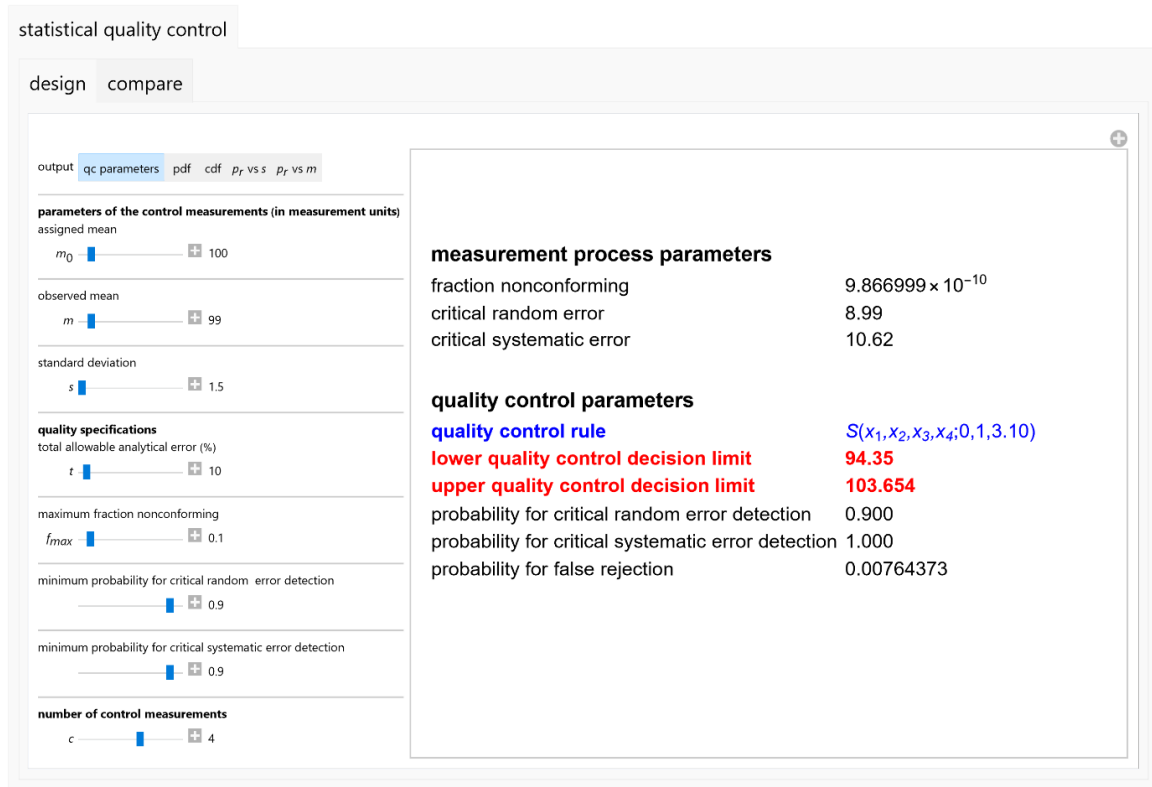


Figure 1: The measurement process parameters and the QC parameters of a measurement process, with the settings at the left.

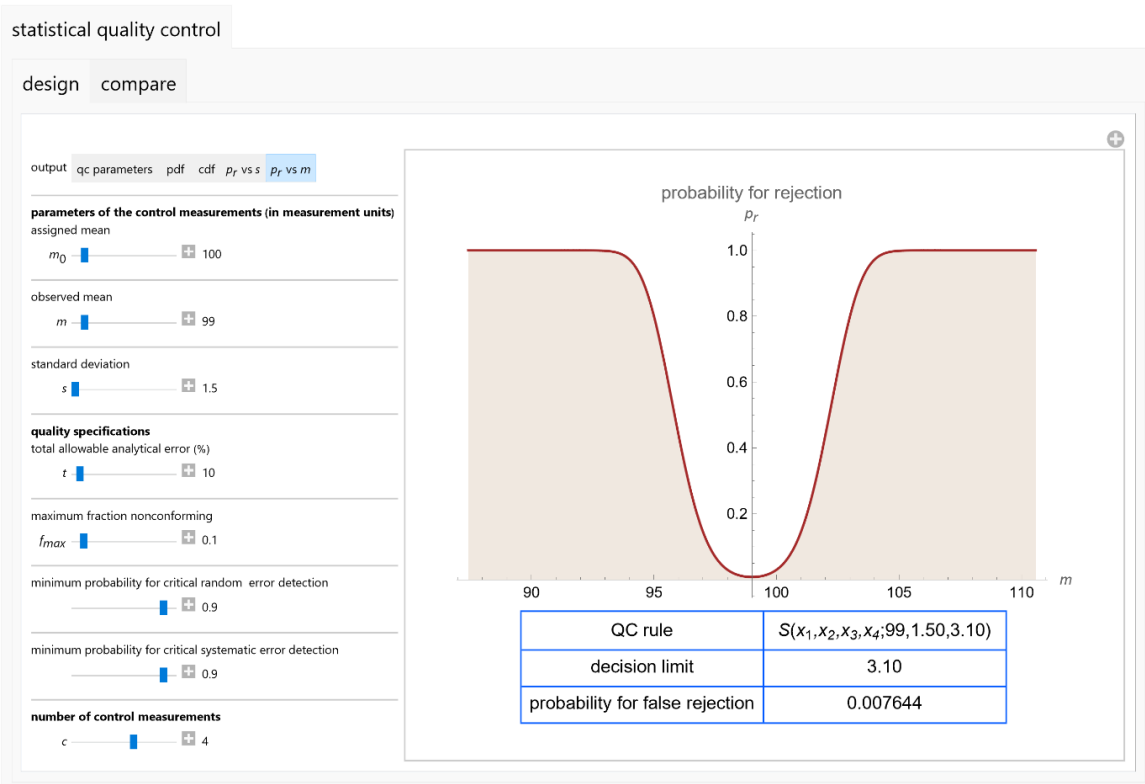


Figure 2: The probability for rejection of a QC rule vs the mean m of the control measurements, with the settings at the left.

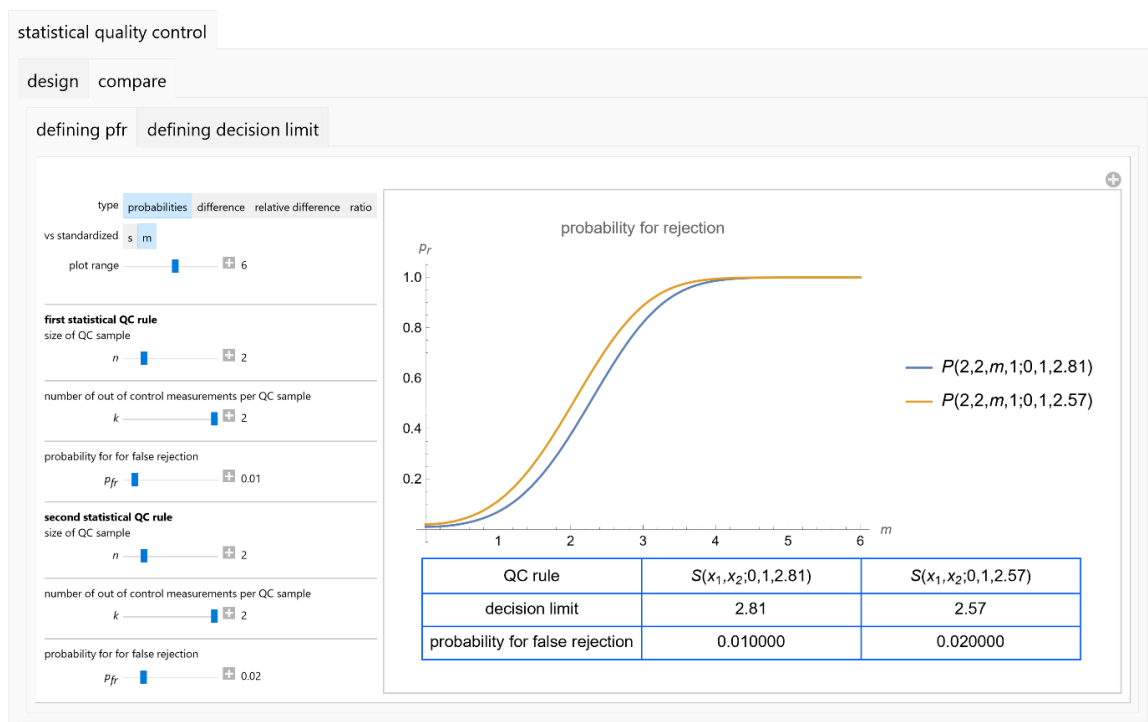


Figure 3: The probabilities for rejection of two alternative QC rules vs the standardized observed mean m of the control measurements, with the settings at the left.

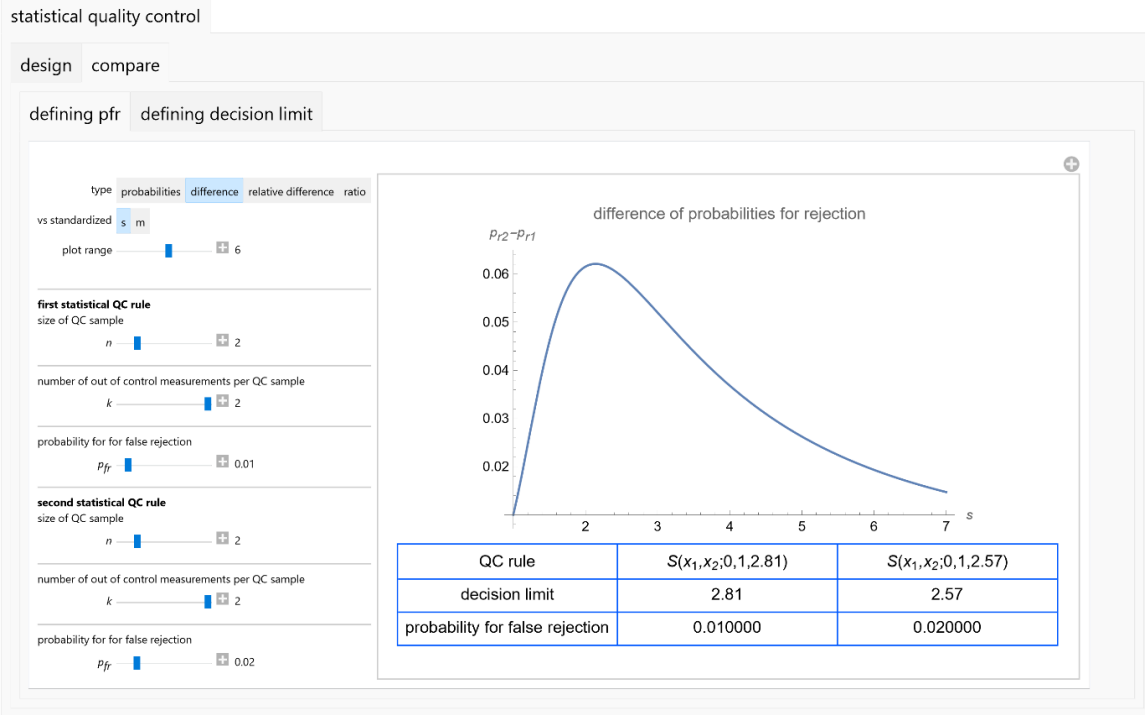


Figure 4: The difference between the probabilities for rejection of two alternative QC rules vs the standardized standard deviation s of the control measurements, with the settings at the left.

Details

Let m_0 , m , and s be the assigned mean, the observed mean, and the standard deviation of the control measurements. Let t be the total allowable analytical error (expressed as a percentage of the assigned mean), and f_{max} the maximum acceptable fraction nonconforming. Then, the following equations are used to estimate the respective parameters [1, 2, 3]:

(a) The fraction nonconforming f :

$$f = 1 - \int_{m_0(1-\frac{t}{100})}^{m_0(1+\frac{t}{100})} \left(\frac{e^{-\frac{(-m+z)^2}{2s^2}}}{s\sqrt{2\pi}} \right) dz$$

(b) the critical random error s_c :

$$1 - \int_{m_0(1-\frac{t}{100})}^{m_0(1+\frac{t}{100})} \left(\frac{e^{-\frac{(-m+z)^2}{2s_c^2}}}{s_c\sqrt{2\pi}} \right) dz = f_{max}$$

(c) the critical systematic error m_c :

$$\int_{m_0(1-\frac{t}{100})}^{m_0(1+\frac{t}{100})} \left(\frac{e^{-\frac{(\epsilon m_c - m + z)^2}{2s^2}}}{s\sqrt{2\pi}} \right) dz = f_{max}$$

where f_{max} the maximum acceptable fraction nonconforming, and $\epsilon = 1$ if $m_0 < m$ and $\epsilon = -1$ otherwise,

(d) the factor l of the decision limits $m \pm l s$ of the QC rule $S(x; m, s, l)$ is the minimum solution of both of the following two equations for the variable l , where p_{s_c} and p_{m_c} the minimum probabilities for critical random and systematic error detection:

$$\left(\int_{m-ls}^{m+ls} \left(\frac{e^{-\frac{(-m+z)^2}{2s_c^2}}}{s_c \sqrt{2\pi}} \right) dz \right)^n = p_{s_c}$$

$$\left(\int_{m-ls}^{m+ls} \left(\frac{e^{-\frac{(\epsilon m_c - m + z)^2}{2s^2}}}{s \sqrt{2\pi}} \right) dz \right)^n = p_{m_c}$$

with ϵ as before in (c),

(e) the probability for false rejection of the QC rule $S(x; m, s, l)$:

$$P(n, n, m, s; m, s, l) = 1 - \left(\int_{m-ls}^{m+ls} \left(\frac{e^{-\frac{(-m+z)^2}{2s^2}}}{s \sqrt{2\pi}} \right) dz \right)^n$$

(f) the probability for rejection for the random error s_e and the systematic error m_e of the QC rule $S(x; m, s, l)$:

$$P(n, n, m_e, s_e; m, s, l) = 1 - \left(\int_{m-ls}^{m+ls} \left(\frac{e^{-\frac{(m_e - m + z)^2}{2s_e^2}}}{s_e \sqrt{2\pi}} \right) dz \right)^n$$

$$P(n, k, m_e, s_e; m, s, l) = P(k, k, m_e, s_e; m, s, l) + P(n - k, n - k, m, s; m, s, l) - P(k, k, m_e, s_e; m, s, l)P(n - k, n - k, m, s; m, s, l)$$

Conclusion

The program *Quality* can be used as an educational or laboratory tool to design and evaluate alternative single-value QC rules for a measurement process.

References

- [1] A. T. Hatjimihail, A Tool for the Design and Evaluation of Alternative Quality Control Procedures, Clinical Chemistry 38, 1992 pp. 204–210.
- [2] A.T. Hatjimihail. Tool for Quality Control Design and Evaluation, Wolfram Demonstrations Project, Champaign: Wolfram Research, Inc., 2010.
- [3] A. T. Hatjimihail, Estimation of the Optimal Statistical Quality Control Sampling Time Intervals Using a Residual Risk Measure, PLoS ONE 4(6), 2009 p. e5770.

Abbreviations

QC: quality control
pfr: probability for false rejection
pdf: probability density function
cdf: cumulative distribution function

Notation

$\mathbf{x}$: n -tuple $(x_1, x_2, \dots, x_n)$ of control measurements
 m_0 : the assigned mean of the control measurements
 m : the observed mean of the control measurements
 s : the observed standard deviation of the control measurements
 m_e : the observed mean of the control measurements when the measurement process is out of control
 s_e : the observed standard deviation of the control measurements when the measurement process is out of control
 s_c : the critical random error

m_c : the critical systematic error

p_r : probability for rejection

p_{fr} : probability for false rejection

p_{sc} : the probability for critical random error detection

p_{mc} : the probability for critical systematic error detection

$S(\mathbf{x}; m, s, l)$: statistical QC rule with decision limits $m \pm ls$, applied to $\mathbf{x}$. The rule rejects an analytical run

if: $\exists x_i \in \mathbf{x}: x_i < m - ls \vee x_i > m + ls$

$\mathcal{N}(m, s^2)$: normal distribution with mean m and variance s^2 .

$P_s(n, k, m_e, s_e; m, s, l)$: probability for rejection of the QC rule $S(\mathbf{x}; m, s, l)$ applied to a n -tuple $\mathbf{x}$ of control measurements, k of them distributed as $\mathcal{N}(m_e, s_e^2)$ and $n - k$ as $\mathcal{N}(m, s^2)$

Interface

1. Design

1.1. Input.

1.1.1. Parameters of the control measurements (in arbitrary measurement units)

1.1.1.1. The assigned mean m_0 ($0.01 \leq m_0 \leq 1000.0$),

1.1.1.2. The observed mean m when the process is in control ($0.01 \leq m \leq 1000.0$), and

1.1.1.3. The standard deviation s when the process is in control ($0.01 \leq s \leq 1000.0$).

1.1.2. Quality specifications of the measurement process

1.1.2.1. The total allowable error t (as a percentage of the assigned mean)
($0.01 \leq t \leq 200.0$),

1.1.2.2. The maximum acceptable fraction f_{max} of measurements nonconforming to the specifications ($0.01 \leq f_{max} \leq 1.0$),

1.1.2.3. The minimum acceptable probability p_{sc} for detection of the critical random error
($0.01 \leq p_{sc} \leq 0.99$).

1.1.2.4. The minimum acceptable probability p_{mc} for detection of the critical systematic error
($0.01 \leq p_{mc} \leq 0.99$).

1.1.2.5. The number n of control measurements ($1 \leq n \leq 6$).

1.2. Output

1.2.1. Table of QC parameters

1.2.1.1. The fraction nonconforming f ,

1.2.1.2. The critical random error s_c ,

1.2.1.3. The critical systematic error m_c ,

1.2.1.4. The QC decision limits $m \pm ls$,

1.2.1.5. The probability p_{sc} for critical random error detection,

1.2.1.6. The probability p_{mc} for critical systematic error detection

1.2.1.7. The probability p_{fr} for false rejection

1.2.2. Plots

1.2.2.1. The probability density function (pdf) plot of the control measurements,

1.2.2.2. The cumulative distribution function (cdf) plot of the control measurements,

1.2.2.3. the probability for rejection plot for the random error (p_r vs s), and

1.2.2.4. the probability for rejection plot for the systematic error (p_r vs m).

The parameters are estimated and the functions are plotted if $10^{-100} \leq f \leq f_{max}$

2. Compare

2.1. Input.

2.1.1. Plot options

2.1.1.1. Type of error:

- 2.1.1.1.1. Random error
- 2.1.1.1.2. Systematic error

2.1.1.2. Type of plot:

- 2.1.1.2.1. The probabilities for rejection of both QC rules
- 2.1.1.2.2. The difference between the probabilities for rejection
- 2.1.1.2.3. The relative difference between the probabilities for rejection
- 2.1.1.2.4. The ratio of the probabilities for rejection

2.1.1.3. Plot range

- 2.1.1.3.1. The plotted range r of the y -axis (1.0 – 10.0)

2.1.2. QC parameters

The following parameters are defined for two alternative QC procedures:

- 2.1.2.1. The number n of the control measurements ($1 \leq n \leq 6$).
- 2.1.2.2. The number k of the control measurements distributed as $\mathcal{N}(m, s^2)$
- 2.1.2.3. Either:
 - 2.1.2.3.1. The probability p_{fr} for false rejection ($0.000001 \leq p_{fr} \leq 0.1$) or
 - 2.1.2.3.2. The QC decision limit l ($1.0 \leq l \leq 4.0$)

2.2. Output

2.2.1. Table of QC parameters

- 2.2.1.1. The QC decision limit l s,
- 2.2.1.2. The probability p_{fr} for false rejection

2.2.2. Plots

- 2.2.2.1. The probabilities for rejection of both QC rules
- 2.2.2.2. The difference between the probabilities for rejection
- 2.2.2.3. The relative difference between the probabilities for rejection
- 2.2.2.4. The ratio of the probabilities for rejection

Availability and requirements

Project name: *Quality*

Version: 1.0.5

DOI: [10.5281/zenodo.20278408](https://doi.org/10.5281/zenodo.20278408)

Project home page: <https://www.hcsl.com/Tools/Quality>

Source notebook: <https://www.hcsl.com/Tools/Quality/Quality.nb>

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Programming language: Wolfram Language

Other software requirements: Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®

System requirements: Intel Core i7 or equivalent CPU and 16 GB+ of RAM

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Citation:

Chatzimichail RA, Hatjimihail AT. *Quality: A Software Tool for Statistical Quality Control Design*. Technical Report XX. Hellenic Complex Systems Laboratory; 2022.
<https://doi.org/10.5281/zenodo.20278487>

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First Published: April 19, 2022

Revised: May 19, 2026